Lecture 9 BGP Implementation and IP Header Quiz

1. Which statement best distinguishes eBGP from iBGP?
A. eBGP is used inside one AS, while iBGP is used between different ASes.
B. eBGP is between routers in different ASes, while iBGP distributes externally learned routes within the same AS.
C. eBGP computes shortest internal paths, while iBGP computes least-cost AS paths.
D. eBGP carries only local attributes, while iBGP carries only destination prefixes.
ANS:

2. What is the main role of an IGP in Lecture 9?
A. To advertise prefixes to neighboring ASes.
B. To find internal routes within the AS.
C. To choose the best MED among providers.
D. To replace BGP entirely.
ANS:

3. What is an egress router for a destination?
A. The router with the smallest IP address in the AS.
B. A border router that can reach the external destination.
C. The first router that receives the packet at Layer 2.
D. A router that only speaks iBGP.
ANS:

4. When sending to an external destination, what does a router do first in the two-table model?
A. Use the IGP table to find the final destination directly.
B. Use ARP to find the neighboring AS.
C. Use the BGP table to find the egress router, then use the IGP table to reach it.
D. Use the MED field as the next hop.
ANS:

5. Why can the simple iBGP implementation become a scaling problem?
A. It requires a checksum for every route.
B. It requires roughly $N \times B$ sessions in the simple model.
C. It forces routers to build new physical links.
D. It requires every router to be a border router.
ANS:

6. What is the idea behind hot potato routing?
A. Keep packets inside your own AS as long as possible.
B. Prefer the route with the largest MED.
C. Prefer the shortest ASPATH only.
D. Get rid of the packet as quickly as possible using the nearest egress router.
ANS:

7. What does MED represent?
A. The size of the BGP routing table.
B. The distance from the announcing router to the destination, where lower is preferred.
C. The number of ASes in the path, where higher is preferred.
D. The TTL of a BGP update.
ANS:

8. Which BGP route attribute is local and used to encode policy preference such as Gao-Rexford ranking?
A. ASPATH.

B. LOCAL PREFERENCE.
C. MED.
D. TTL.
ANS:

9. Which BGP message type announces a new route, updates an old route, or withdraws a route?
A. Open.
B. KeepAlive.
C. Notification.
D. Update.
ANS:

10. In the import policy order, which rule is considered first?
A. Shortest ASPATH.
B. Lowest router IP address.
C. Gao-Rexford preference through highest LOCAL PREF.
D. Lowest MED.
ANS:

11. Which IPv4 header field tells the destination which Layer 4 protocol should process the payload?
A. Total Length.
B. Protocol.
C. Header Length.
D. Identification.
ANS:

12. What is the main purpose of TTL in IPv4?
A. To indicate the packet priority.
B. To identify all fragments of one packet.
C. To prevent indefinite forwarding loops by limiting hop count.
D. To measure propagation delay.
ANS:

13. In IPv4 fragmentation, which statement is correct?
A. Each fragment gets a different ID.
B. Fragment offsets are measured in bytes.
C. All fragments of the same packet share an ID, and offsets are measured in multiples of 8 bytes.
D. Fragmentation is done only by the destination host.
ANS:

14. Which set of changes is part of the IPv6 cleanup described in Lecture 9?
A. Add checksums, expand options, and keep router fragmentation.
B. Remove checksums, remove router fragmentation, and remove options in favor of next-header processing.
C. Reduce address size from 32 bits to 16 bits.
D. Replace hop limit with MED.
ANS:

15. Which security problem is associated with the source IP address field?
A. Traceroute.
B. Spoofing.
C. Flow labeling.
D. Header compression.
ANS: